

Threat Detection and Incident Response

Using Wireshark, pfSense, and Wazuh

SOC Analysis Capstone Report

Field	Details
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Submission Date	14 June 2026

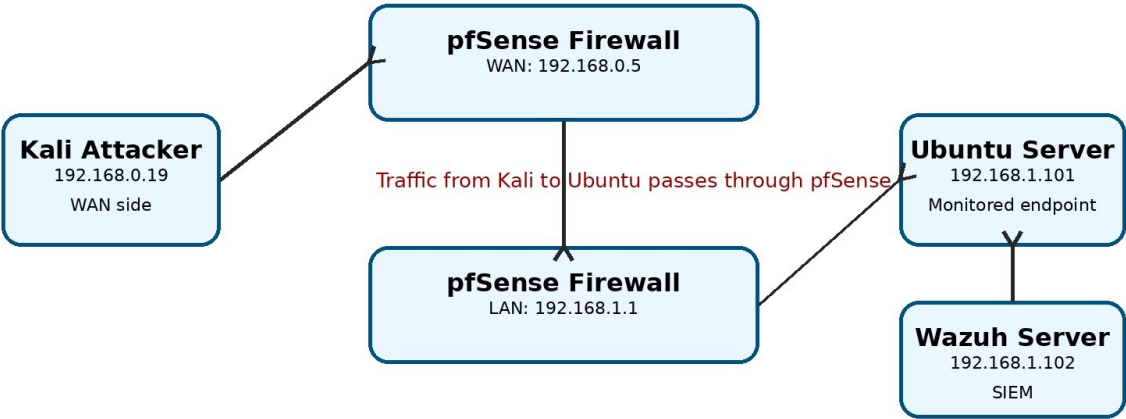


Figure 1. Lab topology used for the Barclays Telco SOC investigation.

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1. Executive Summary

Barclays Telco observed suspicious activity involving an internal Ubuntu server. The activity included host discovery, port scanning, SSH service identification, and repeated SSH authentication attempts from an external host. The investigation was performed from the perspective of a Security Operations Center analyst using Wireshark, pfSense, and Wazuh to collect and correlate evidence.

The investigation found that Kali Linux at 192.168.0.19 performed reconnaissance and SSH access attempts against the Ubuntu server at 192.168.1.101. Wireshark confirmed TCP scanning and SSH traffic, pfSense confirmed firewall activity involving TCP port 22 and other scanned ports, and Wazuh confirmed failed SSH authentication events from the Ubuntu endpoint. A pfSense block rule was implemented to prevent further SSH access from the suspicious Kali host.

2. Project Introduction

This capstone project simulated a SOC investigation for Barclays Telco, a telecommunications organization providing connectivity, voice, and enterprise communication services. The scenario involved unusual activity against an internal Ubuntu server, including possible host discovery, repeated connection attempts, and suspicious SSH login activity from an external host.

My role was to investigate the activity as a SOC analyst, analyze evidence from multiple security tools, apply basic prevention controls, and produce a SOC-style incident report. The lab environment was configured so that Kali acted as the suspicious external host, pfSense acted as the firewall and gateway, Ubuntu acted as the internal monitored endpoint, and Wazuh acted as the SIEM/endpoint monitoring platform.

Asset	Role	IP Address
Kali Linux	Suspicious external host / attacker simulation	192.168.0.19
Ubuntu Server	Internal Barclays Telco server / monitored endpoint	192.168.1.101
pfSense WAN	Firewall WAN interface	192.168.0.5
pfSense LAN	Firewall LAN interface	192.168.1.1
Wazuh Server	SIEM and endpoint monitoring platform	192.168.1.102

```
pfSense 2.7.2-RELEASE amd64 20231206-2010
Bootup complete

FreeBSD/amd64 (pfSense.home.arpa) (ttyv0)

VirtualBox Virtual Machine - Netgate Device ID: 663df41ff62b8c5fe886

*** Welcome to pfSense 2.7.2-RELEASE (amd64) on pfSense ***

WAN (wan)      -> em0      -> v4/DHCP4: 192.168.0.5/24
                                   v6/DHCP6: 2a02:c7c:c23c:a200:a00:27ff:fe60:658
8/64
LAN (lan)      -> em1      -> v4: 192.168.1.1/24

0) Logout (SSH only)          9) pfTop
1) Assign Interfaces          10) Filter Logs
2) Set interface(s) IP address 11) Restart webConfigurator
3) Reset webConfigurator password 12) PHP shell + pfSense tools
4) Reset to factory defaults  13) Update from console
5) Reboot system              14) Enable Secure Shell (sshd)
6) Halt system                 15) Restore recent configuration
7) Ping host                   16) Restart PHP-FPM
8) Shell

Enter an option: █
```

Figure 2. pfSense console showing WAN 192.168.0.5 and LAN 192.168.1.1 interface configuration.

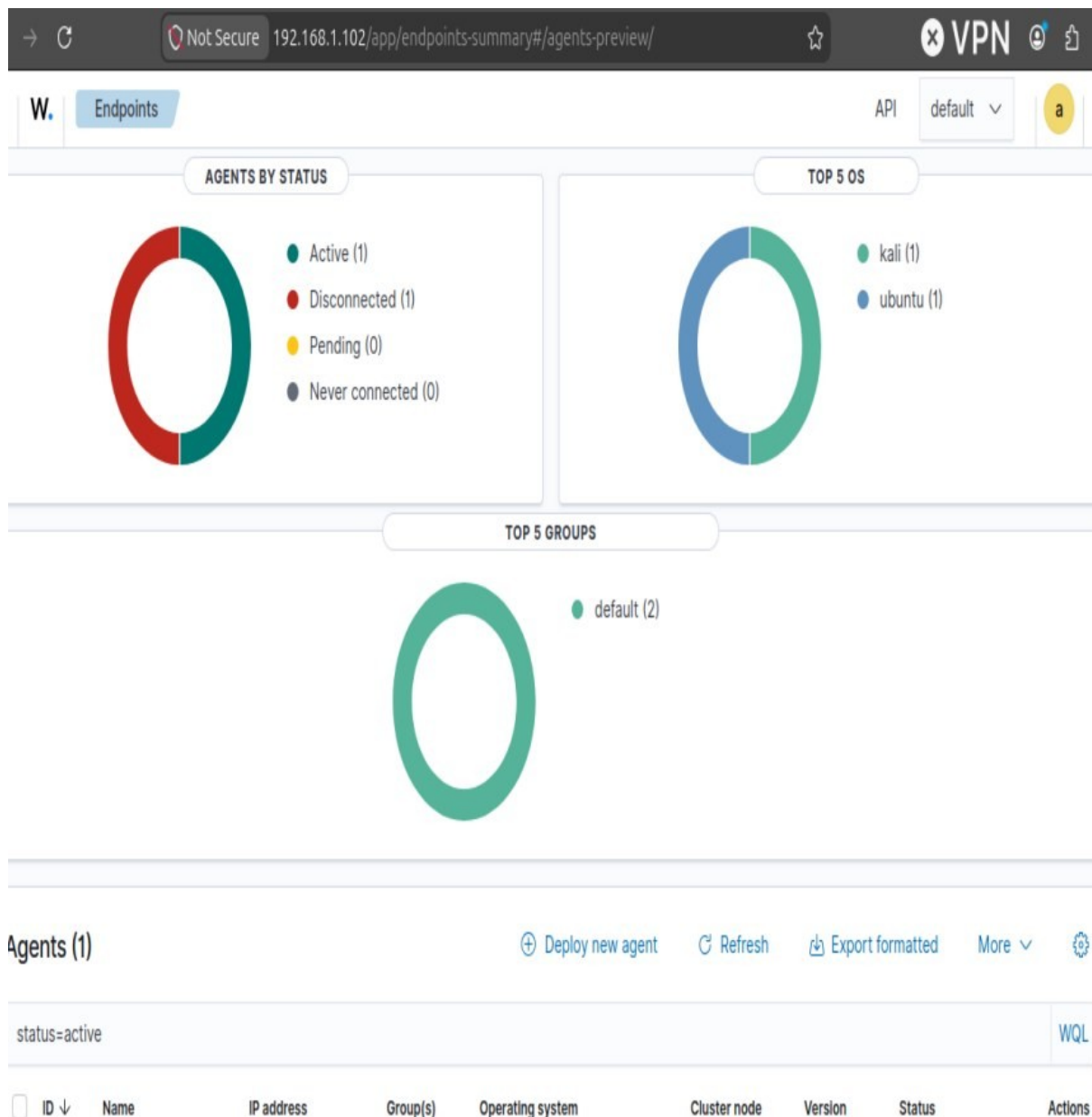


Figure 3. Wazuh dashboard showing one active endpoint agent after the Ubuntu agent was repaired and reconnected.

A static route was required on Kali so that traffic destined for the Ubuntu LAN network would be sent through the pfSense WAN interface. The route used was 192.168.1.0/24 via 192.168.0.5. This routing step was important because firewall rules alone do not guarantee connectivity if the source host does not know the correct path to the target network.

3. Tools Used

Tool	Purpose in the Investigation
Wireshark	Captured and analyzed normal and suspicious

	network traffic, including DNS, HTTP, TCP SYN scan activity, and SSH traffic.
pfSense	Reviewed firewall logs, confirmed allowed/blocked traffic, created aliases, and applied SSH prevention controls.
Wazuh	Monitored the Ubuntu endpoint and generated alerts for failed SSH authentication and invalid user login attempts.
Kali Linux	Simulated the suspicious external host used to perform host discovery, port scanning, and SSH login attempts.
Ubuntu	Target system and monitored endpoint representing Barclays Telco internal server.
Nmap	Performed host discovery, basic port scanning, and targeted SSH service scanning.
SSH	Remote access service used to simulate authentication attempts.

4. Timeline of Events

Approx. Time	Event	Tool(s)	Evidence
Phase 1	Lab topology validated; Kali placed on WAN side and Ubuntu on LAN side; Ubuntu agent confirmed active in Wazuh.	pfSense, Wazuh, Linux CLI	IP inventory, pfSense interface view, Wazuh agent status
Phase 2	Normal traffic baseline captured, including DNS, HTTP, ICMP, and SSH.	Wireshark, Kali	Baseline packet captures and screenshots
Jun 13 17:59	Ping sweep performed from Kali	Nmap, Wireshark	Nmap ping sweep

	against 192.168.1.0/24.		screenshot
Jun 13 18:08	Basic port scan performed against Ubuntu.	Nmap, Wireshark, pfSense	Nmap output showing 22/tcp open ssh
Jun 13 18:21-18:29	SSH service scan and failed SSH login attempts generated from Kali.	Nmap, SSH, Wazuh	SSH scan and failed login screenshots
Jun 13 18:23-18:27	Wazuh alerts observed for failed SSH authentication and invalid user attempts.	Wazuh	Wazuh Threat Hunting alerts and alert details
Phase 5	pfSense firewall logs reviewed and SSH block rule created using suspicious host alias.	pfSense	Allowed traffic logs, alias, block rule, blocked traffic logs

5. Summary of Findings

5.1 Wireshark Findings

Wireshark was used first to establish a baseline of normal traffic. Normal DNS queries, HTTP web requests, and successful SSH traffic were captured. This baseline helped show what expected activity looked like before suspicious activity was generated.

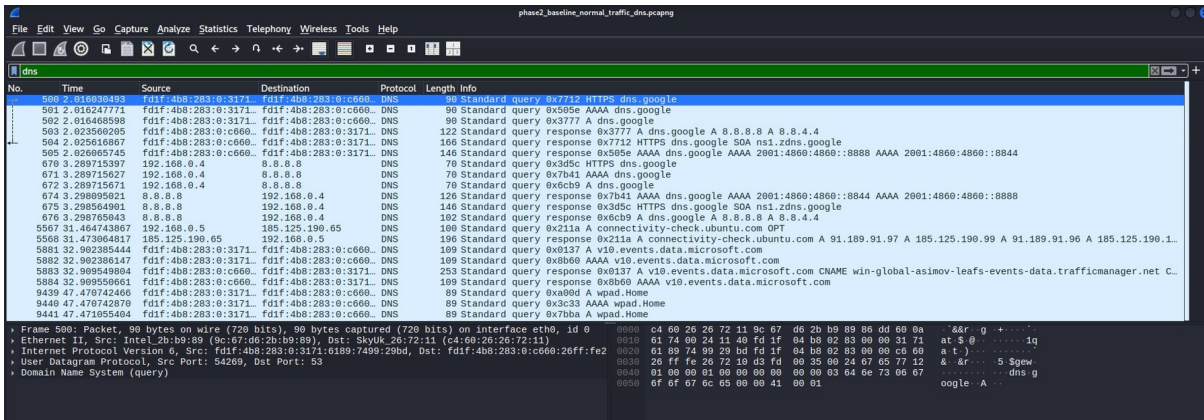


Figure 4. Baseline DNS traffic showing standard DNS queries and responses.

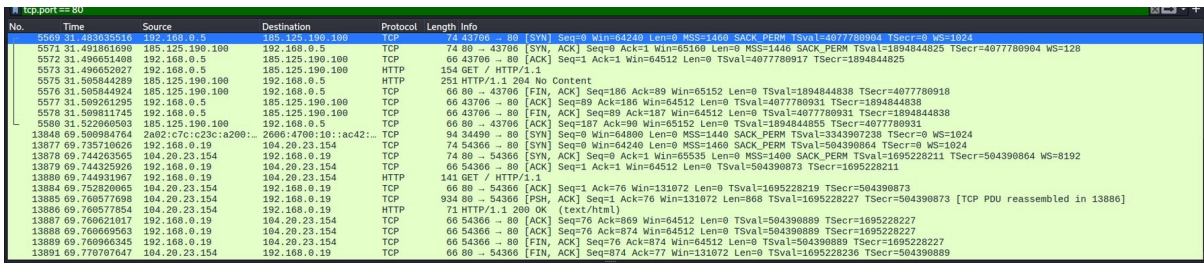


Figure 5. Baseline HTTP traffic using tcp.port == 80, including web requests and responses.

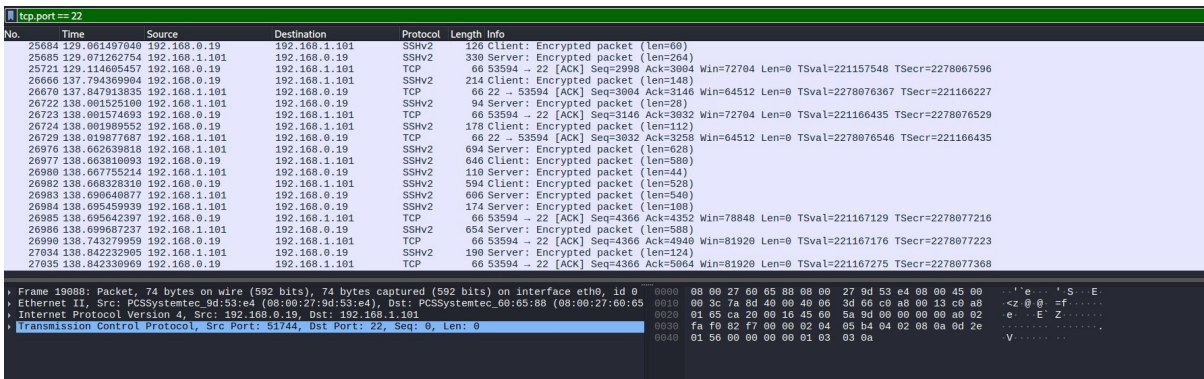


Figure 6. Successful baseline SSH traffic showing SSHv2 encrypted packets between Kali and Ubuntu.

During suspicious activity analysis, Wireshark showed multiple TCP SYN packets from Kali to Ubuntu across many destination ports. This pattern is consistent with port scanning because the source host attempted to initiate connections to multiple services on the target in a short period of time.

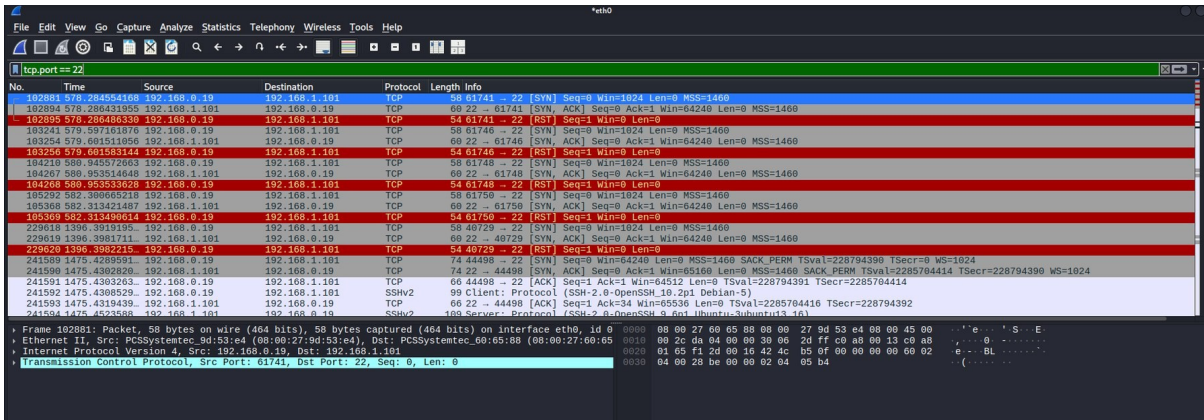


Figure 7. Wireshark SSH traffic evidence showing TCP port 22 and SSHv2 traffic between Kali and Ubuntu.

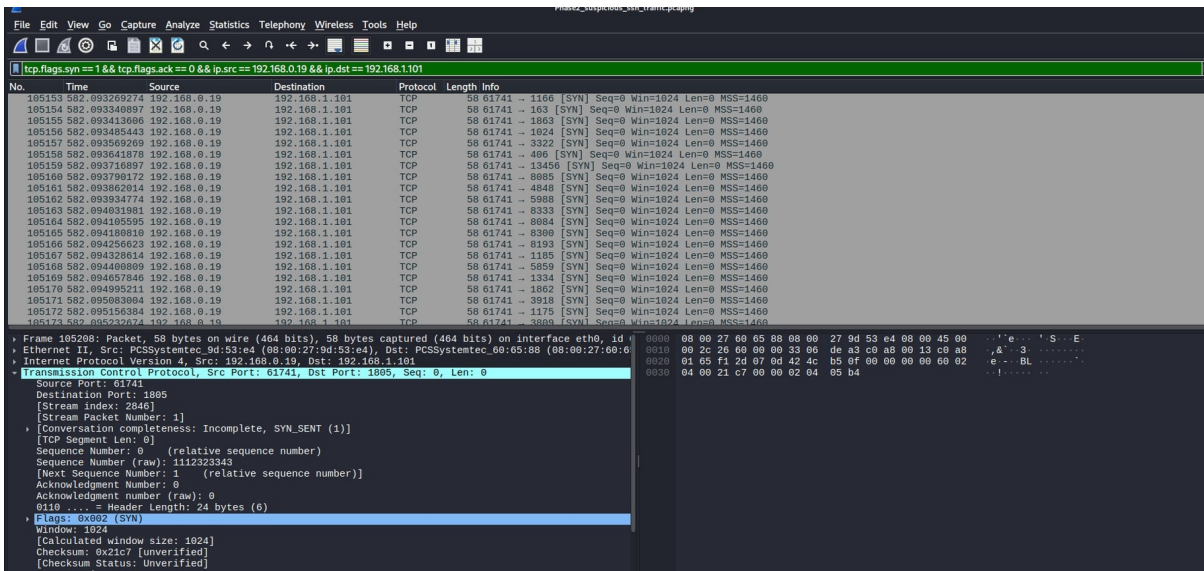


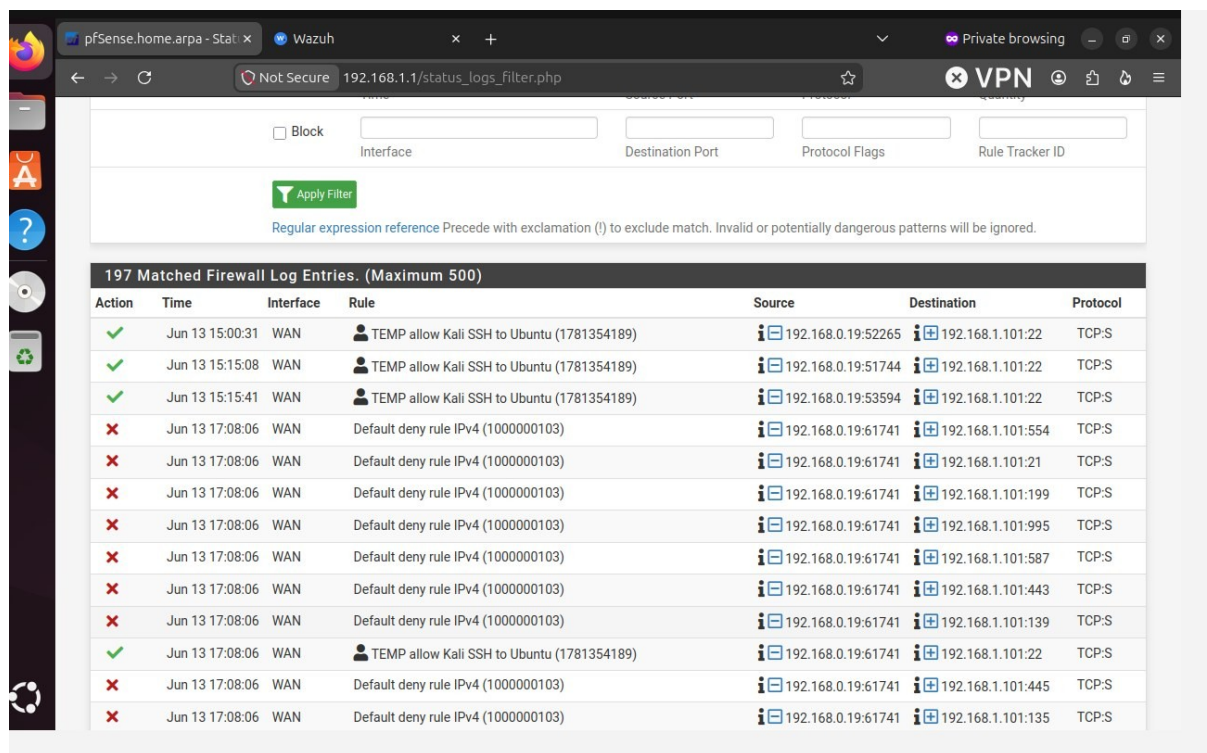
Figure 8. Wireshark scan evidence showing multiple TCP SYN packets from Kali to Ubuntu on different destination ports.

Important Wireshark filters used:

- dns
- tcp.port == 80
- tcp.port == 22
- tcp.flags.syn == 1 && tcp.flags.ack == 0 && ip.src == 192.168.0.19 && ip.dst == 192.168.1.101
- ip.addr == 192.168.0.19 && ip.addr == 192.168.1.101

5.2 pfSense Findings

pfSense firewall logs confirmed traffic from Kali to Ubuntu. SSH traffic to destination TCP port 22 was allowed by the temporary SSH allow rule. The same logs also showed that other scanned ports were blocked by the default deny rule, which supported the Nmap and Wireshark evidence of scanning behavior.



The screenshot displays the pfSense firewall log interface. At the top, there is a search bar with fields for Block, Interface, Destination Port, Protocol Flags, and Rule Tracker ID. Below the search bar is a green 'Apply Filter' button and a note about regular expression references. The main section shows '197 Matched Firewall Log Entries. (Maximum 500)'. The log entries are as follows:

Action	Time	Interface	Rule	Source	Destination	Protocol
✓	Jun 13 15:00:31	WAN	TEMP allow Kali SSH to Ubuntu (1781354189)	192.168.0.19:52265	192.168.1.101:22	TCP:S
✓	Jun 13 15:15:08	WAN	TEMP allow Kali SSH to Ubuntu (1781354189)	192.168.0.19:51744	192.168.1.101:22	TCP:S
✓	Jun 13 15:15:41	WAN	TEMP allow Kali SSH to Ubuntu (1781354189)	192.168.0.19:53594	192.168.1.101:22	TCP:S
✗	Jun 13 17:08:06	WAN	Default deny rule IPv4 (1000000103)	192.168.0.19:61741	192.168.1.101:554	TCP:S
✗	Jun 13 17:08:06	WAN	Default deny rule IPv4 (1000000103)	192.168.0.19:61741	192.168.1.101:21	TCP:S
✗	Jun 13 17:08:06	WAN	Default deny rule IPv4 (1000000103)	192.168.0.19:61741	192.168.1.101:199	TCP:S
✗	Jun 13 17:08:06	WAN	Default deny rule IPv4 (1000000103)	192.168.0.19:61741	192.168.1.101:995	TCP:S
✗	Jun 13 17:08:06	WAN	Default deny rule IPv4 (1000000103)	192.168.0.19:61741	192.168.1.101:587	TCP:S
✗	Jun 13 17:08:06	WAN	Default deny rule IPv4 (1000000103)	192.168.0.19:61741	192.168.1.101:443	TCP:S
✗	Jun 13 17:08:06	WAN	Default deny rule IPv4 (1000000103)	192.168.0.19:61741	192.168.1.101:139	TCP:S
✓	Jun 13 17:08:06	WAN	TEMP allow Kali SSH to Ubuntu (1781354189)	192.168.0.19:61741	192.168.1.101:22	TCP:S
✗	Jun 13 17:08:06	WAN	Default deny rule IPv4 (1000000103)	192.168.0.19:61741	192.168.1.101:445	TCP:S
✗	Jun 13 17:08:06	WAN	Default deny rule IPv4 (1000000103)	192.168.0.19:61741	192.168.1.101:135	TCP:S

Figure 9. pfSense firewall logs showing allowed SSH traffic to Ubuntu and blocked scan attempts to other ports.

A pfSense alias named Suspicious_Kali_Host was created for 192.168.0.19 to make firewall rules easier to manage. A block rule was then created on the WAN interface to block TCP port 22 traffic from the alias to the Ubuntu server. The block rule was placed above the previous allow rule because pfSense processes firewall rules from top to bottom.

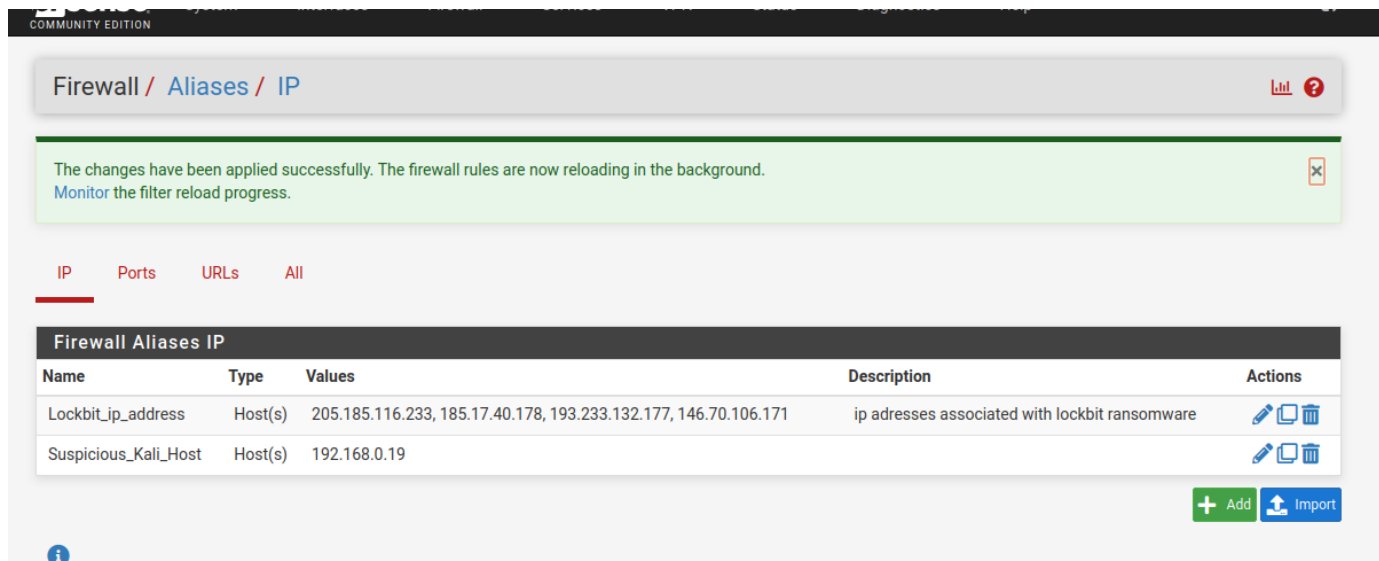


Figure10: pfSense alias configuration for Suspicious_Kali_Host = 192.168.0.19

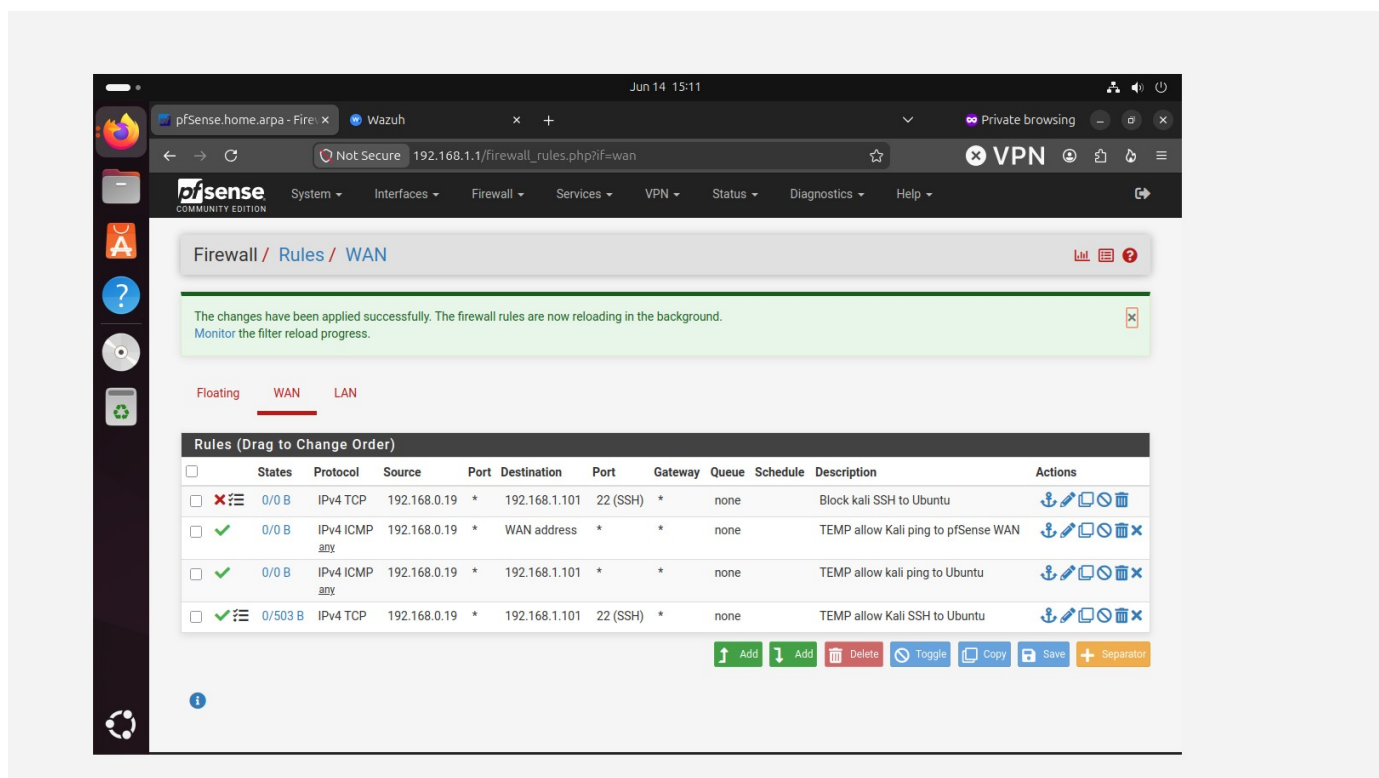


Figure 11: pfSense WAN block rule: block Suspicious_Kali_Host to Ubuntu 192.168.1.101 on SSH/22

```
(tomi@kali)-[~]
$ nc -vz -n 192.168.1.101 22
(UNKNOWN) [192.168.1.101] 22 (ssh) open

(tomi@kali)-[~]
$ nc -vz -n 192.168.1.101 22

(UNKNOWN) [192.168.1.101] 22 (ssh) : Connection timed out

(tomi@kali)-[~]
$

(tomi@kali)-[~]
$

(tomi@kali)-[~]
$

(tomi@kali)-[~]
$ nmap -Pn -n -p 22 192.168.1.101
Starting Nmap 7.98 ( https://nmap.org ) at 2026-06-14 15:15 +0100
Nmap scan report for 192.168.1.101
Host is up.

PORT      STATE      SERVICE
22/tcp    filtered  ssh

Nmap done: 1 IP address (1 host up) scanned in 2.15 seconds

(tomi@kali)-[~]
$
```

Figure 12: Kali test showing SSH blocked after the pfSense prevention rule

✗	Jun 14 14:15:56	WAN	Default deny rule IPv6 (1000000105)	📡 [fe80::c660:26ff:fe26:7211]:47246	📡 [fe80::a00:27ff:fe60:6588]:546	UDP
✗	Jun 14 14:15:58	LAN	Default deny rule IPv6 (1000000105)	📡 [fe80::a00:27ff:fe1c:e782]:41144	📡 [2620:2d:4000:1::1099]:80	TCP:S
✗	Jun 14 14:15:59	WAN	👤 Block kali SSH to Ubuntu (1781446207)	📡 192.168.0.19:34959	📡 192.168.1.101:22	TCP:S
✗	Jun 14 14:15:59	LAN	Default deny rule IPv6 (1000000105)	📡 [fe80::a00:27ff:fe1c:e782]:41144	📡 [2620:2d:4000:1::1099]:80	TCP:S
✗	Jun 14 14:15:59	WAN	Default deny rule IPv4 (1000000103)	📡 192.168.0.9:6537	📡 255.255.255.255:6537	UDP
✗	Jun 14 14:15:59	WAN	Default deny rule IPv4 (1000000103)	📡 192.168.0.9:6537	📡 255.255.255.255:6537	UDP
✗	Jun 14 14:16:00	WAN	👤 Block kali SSH to Ubuntu (1781446207)	📡 192.168.0.19:34961	📡 192.168.1.101:22	TCP:S
✗	Jun 14 14:16:00	LAN	Default deny rule IPv6 (1000000105)	📡 [fe80::a00:27ff:fe1c:e782]:41144	📡 [2620:2d:4000:1::1099]:80	TCP:S

Figure 12: pfSense firewall logs showing blocked SSH traffic from 192.168.0.19 to 192.168.1.101:22

5.3 Wazuh Findings

Wazuh confirmed endpoint-level evidence from the Ubuntu server. The Ubuntu agent was active, which confirmed that Wazuh could receive endpoint logs. Threat Hunting searches for SSH, failed password, and authentication failure returned alerts related to failed logins and invalid users.

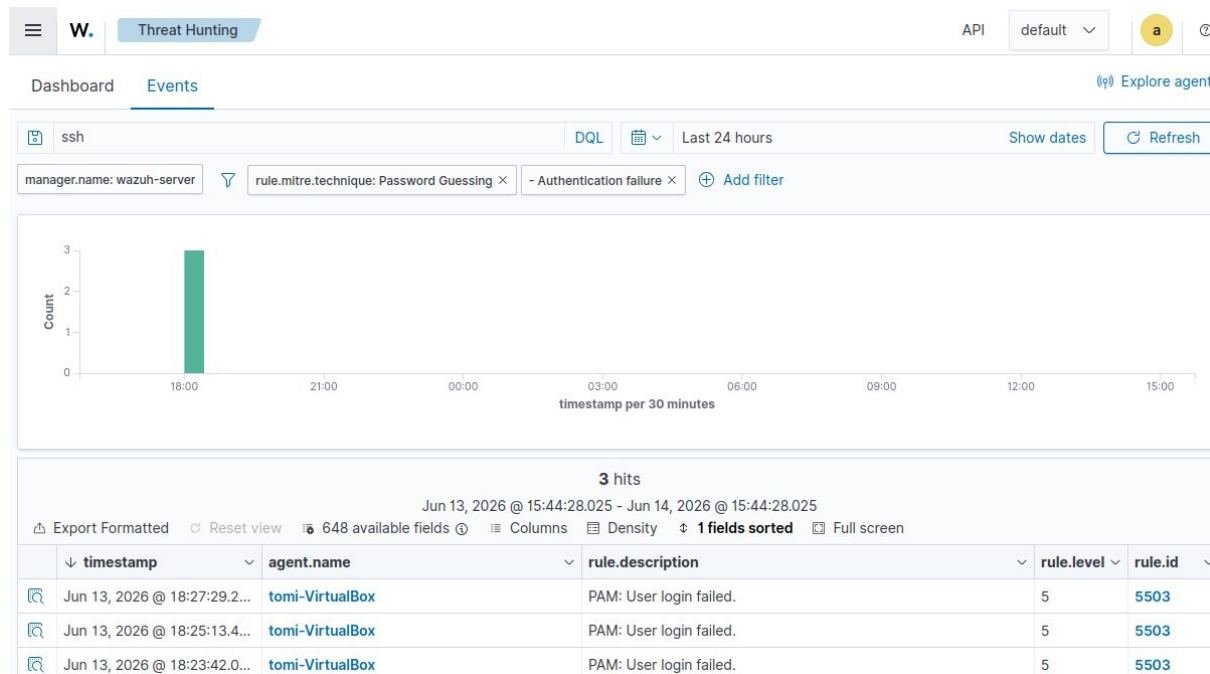


Figure 10. Wazuh Threat Hunting results showing PAM user login failed alerts.

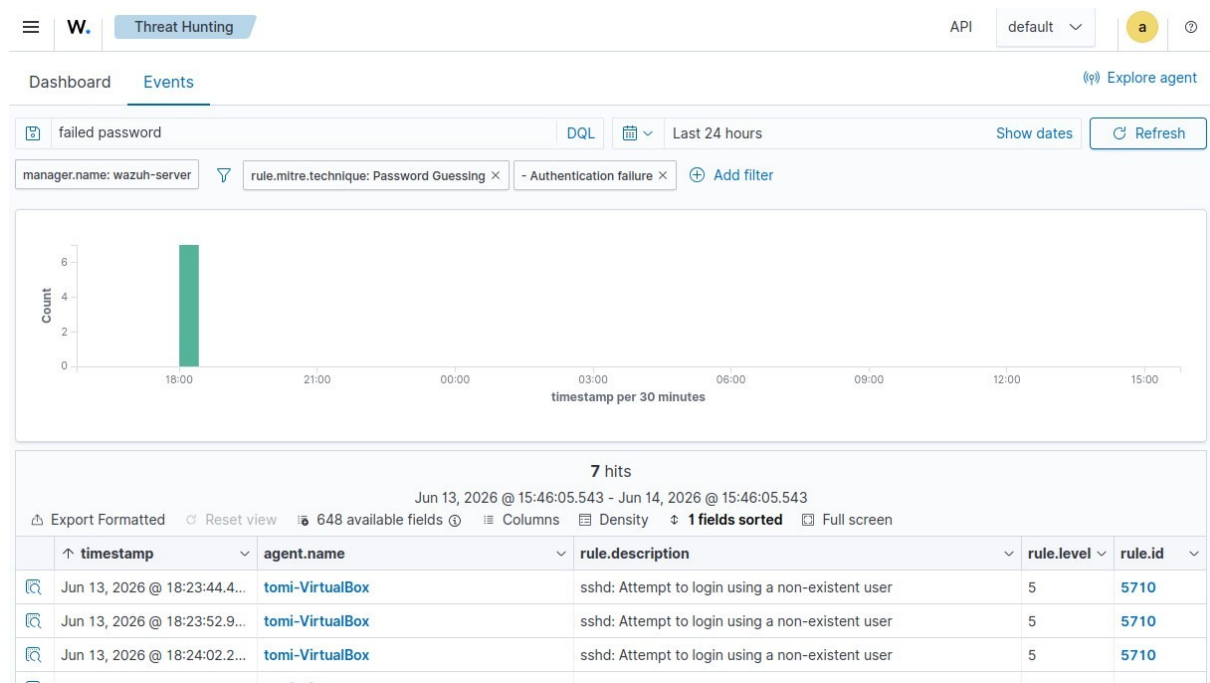


Figure 11. Wazuh Threat Hunting results showing non-existent SSH user login attempts.

W.	Discover	wazuh-alerts-4.x-2026.06.13#R5ECwp4BPSA_gy-01_Cp	a
Table	JSON		
@timestamp		Jun 13, 2026 @ 18:23:44.415	
_index		wazuh-alerts-4.x-2026.06.13	
agent.id		002	
agent.ip		192.168.1.101	
agent.name		tomi-VirtualBox	
data.srcip		192.168.0.19	
data.srcuser		fakeuser	
decoder.name		sshd	
decoder.parent		sshd	
full_log		Jun 13 17:23:43 tomi-VirtualBox sshd[15117]: Failed password for invalid user fakeuser from 192.168.0.19 port 44498 ssh2	
id		1781371424.3742285	
input.type		log	
location		journald	
manager.name		wazuh-server	
predecoder.hostname		tomi-VirtualBox	
predecoder.program_name		sshd	
predecoder.timestamp		Jun 13 17:23:43	
rule.description		sshd: Attempt to login using a non-existent user	

Figure 12. Wazuh alert detail showing invalid user fakeuser from Kali at 192.168.0.19.

W.	Discover	wazuh-alerts-4.x-2026.06.13#R5ECwp4BPSA_gy-01_Cp
predecoder.hostname		tomi-VirtualBox
predecoder.program_name		sshd
predecoder.timestamp		Jun 13 17:23:43
rule.description		sshd: Attempt to login using a non-existent user
# rule.firedtimes		2
rule.gdpr		IV_35.7.d, IV_32.2
rule.gpg13		7.1
rule.groups		syslog, sshd, authentication_failed, invalid_login
rule.hipaa		164.312.b
rule.id		5710
# rule.level		5
rule.mail		false
rule.mitre.id		T1110.001, T1021.004
rule.mitre.tactic		Credential Access, Lateral Movement
rule.mitre.technique		Password Guessing, SSH
rule.nist_800_53		AU.14, AC.7, AU.6
rule.pci_dss		10.2.4, 10.2.5, 10.6.1
rule.tsc		CC6.1, CC6.8, CC7.2, CC7.3
timestamp		Jun 13, 2026 @ 18:23:44.415

Figure 13. Wazuh rule metadata and MITRE mapping to Password Guessing and SSH.

One alert showed a failed password attempt for invalid user fakeuser from source IP 192.168.0.19. Wazuh classified the event as “sshd: Attempt to login using a non-existent user”

with rule ID 5710 and rule level 5. The alert was mapped to MITRE ATT&CK tactics Credential Access and Lateral Movement, with techniques including Password Guessing and SSH.

6. Incident Assessment

Assessment Area	Finding
Source IP	192.168.0.19 (Kali suspicious external host)
Target System	Ubuntu server / tomi-VirtualBox at 192.168.1.101
Suspicious Activity	Host discovery, TCP port scan, SSH service scan, failed SSH login attempts using invalid users
Firewall Evidence	pfSense logs showed allowed SSH traffic to 192.168.1.101:22 and blocked scan traffic to other ports; SSH block rule later applied.
Endpoint Evidence	Wazuh detected failed SSH authentication and invalid user activity from 192.168.0.19.
Final Assessment	Suspicious activity consistent with reconnaissance followed by SSH password guessing attempts.

The activity was assessed as suspicious because the external Kali host performed host discovery and scanning before attempting SSH access against the internal Ubuntu server. Multiple failed login attempts and invalid usernames were observed. While this activity was generated intentionally in a controlled lab, the pattern is consistent with real-world reconnaissance and password guessing behavior.

7. Recommendations

- Restrict SSH access to trusted administrative IP addresses only.
- Keep the pfSense block rule for suspicious external SSH traffic and use aliases for easier firewall rule management.
- Disable SSH root login and consider disabling password-based SSH authentication in favor of key-based authentication.
- Implement strong password policies and multi-factor authentication where possible.

- Deploy Fail2Ban or equivalent controls to temporarily block hosts after repeated SSH failures.
- Create Wazuh alert rules or dashboards for repeated failed SSH logins, invalid users, and successful login after multiple failures.
- Review pfSense logs regularly and ensure firewall rules are ordered correctly, with deny rules above broad allow rules where appropriate.
- Maintain updated systems and monitor Ubuntu authentication logs for unusual login patterns.

8. Conclusion

This project demonstrated how a SOC analyst can investigate suspicious activity using multiple sources of evidence. Wireshark provided packet-level visibility into scans and SSH traffic, pfSense provided firewall-level evidence and prevention capability, and Wazuh provided endpoint-level confirmation of failed SSH authentication. By correlating these tools, the investigation showed how reconnaissance and SSH password guessing activity can be detected, validated, and controlled in a structured SOC workflow.

9. References

- SOC Analysis Capstone Project Workflow, Barclays Telco case study.
- Final SOC Capstone Report Sample Template.
- Wireshark packet capture evidence collected during the lab.
- pfSense firewall logs and firewall rule screenshots collected during the lab.
- Wazuh Threat Hunting alerts and alert details collected during the lab.
- MITRE ATT&CK mappings shown in Wazuh alert details.

10. Appendices: Evidence and Screenshot Checklist

The following screenshot checklist shows which evidence images should be included in the final submission. Items marked as inserted are already included in this draft. Items marked as required placeholder should be added from your saved screenshots before final submission.

Section	Screenshot Evidence to Add /	Status / Suggested Filename
Phase 1	IP address confirmation / pfSense WAN and	Inserted - phase1_pfsense_interfaces.png

	LAN interface view	
Phase 1	Ubuntu agent active in Wazuh	Inserted - phase1_wazuh_ubuntu_agent_active.png
Phase 2	Normal DNS traffic in Wireshark	Inserted - phase2_normal_dns_lookup.png
Phase 2	Normal HTTP traffic in Wireshark	Inserted - phase2_normal_http_tcp_port_80.png
Phase 2	Normal SSH traffic in Wireshark	Inserted - phase2_normal_ssh_successful_connection.png
Phase 3	Nmap ping sweep output	Inserted - phase3_nmap_ping_sweep.png
Phase 3	Nmap basic port scan showing 22/tcp open ssh	Inserted - phase3_nmap_basic_port_scan.png
Phase 3	SSH service scan and failed SSH attempts	Inserted - phase3_failed_ssh_login_attempts.png
Phase 4	Wireshark SSH traffic on TCP port 22	Inserted - phase4_wireshark_ssh_traffic_analysis.png
Phase 4	Wireshark TCP SYN scan evidence	Inserted - phase4_wireshark_tcp_scan_kali_to_ubuntu.png
Phase 5	pfSense logs showing Kali-to-Ubuntu SSH traffic	Inserted - phase5_pfsense_ssh_allowed_and_scan_blocked_logs.png
Phase 5	pfSense alias configuration	Inserted- phase5_pfsense_alias_suspicious_kali_host.png
Phase 5	pfSense SSH block rule	Inserted- phase5_pfsense_ssh_block_rule.png
Phase 5	Block rule above allow rule	Inserted- phase5_pfsense_block_rule_above_allow_rule.png
Phase 5	Kali test showing SSH blocked	Inserted - phase5_kali_ssh_block_test.png
Phase 5	pfSense blocked	Inserted- phase5_pfsense_blocked_ssh_logs.png

	SSH log entry	
Phase 6	Wazuh SSH alert list	Inserted - phase6_wazuh_ssh_failed_login_alerts.png
Phase 6	Wazuh alert details for invalid user	Inserted - phase6_wazuh_ssh_alert_details_invalid_user.png
Phase 6	Wazuh MITRE mapping/password guessing	Inserted - phase6_wazuh_mitre_password_guessing.png

Appendix A: Wireshark Evidence Notes

Filters used: dns, tcp.port == 80, tcp.port == 22, and tcp.flags.syn == 1 && tcp.flags.ack == 0 && ip.src == 192.168.0.19 && ip.dst == 192.168.1.101.

Appendix B: pfSense Evidence Notes

Evidence include allowed SSH logs, SSH block rule, alias configuration, rule order, and blocked SSH logs.

Appendix C: Wazuh Evidence Notes

Evidence include Ubuntu agent active status, SSH failed login alerts, invalid user alert details, source IP, attempted username, rule level, and MITRE mapping.